Unified Engineering Workbench

PASS Science Works · R&D Internship 2026–2027 · Prototype Technical Report

1. Scope and architecture

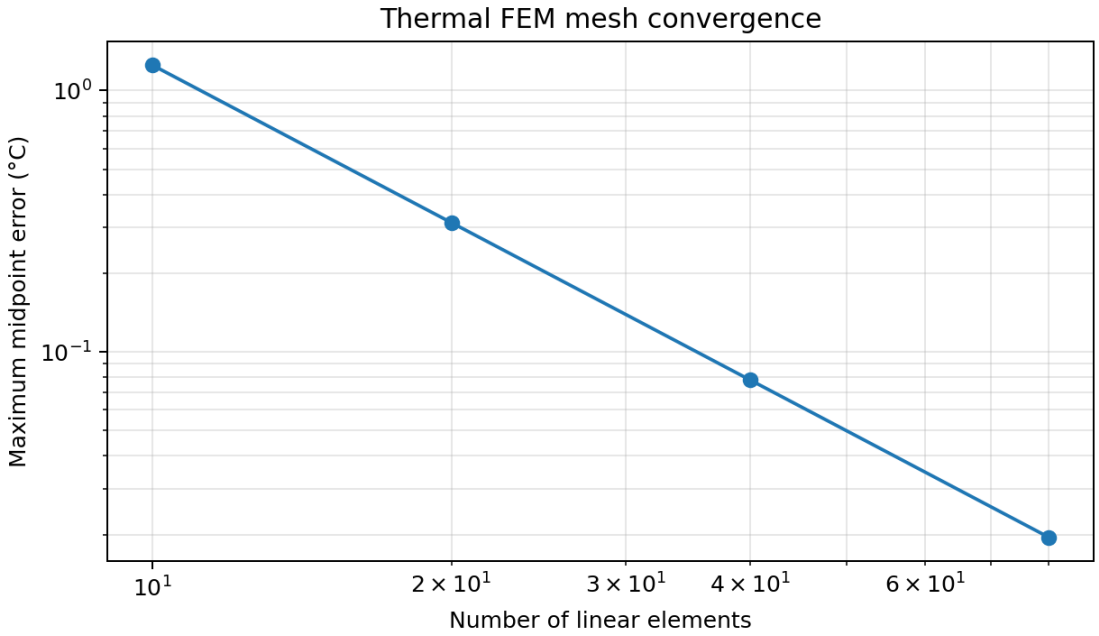
The prototype is organized as a C++17 core with separate modules for thermal FEM, geometry parsing, multicore benchmarking, embedded-device mocks and backpropagation. CMake builds a command-line executable and CTest test target. A separate dark dashboard concept is included in frontend/. The dashboard is not yet connected to the C++ executable, so the deliverable should be treated as a working computational prototype rather than a fully integrated GUI application.

2. Thermal finite element analysis

A uniform rod is modelled with two-node linear elements. For element length h, the stiffness matrix is (kA/h)[[1,-1],[-1,1]] and the consistent load vector is (qAh/2)[1,1]. Element contributions are assembled into a tridiagonal global system. Prescribed temperatures T(0)=100 °C and T(L)=20 °C are imposed by eliminating boundary couplings and replacing boundary equations. The analytical solution is T(x)=100−80x/L+[q/(2k)]x(L−x).

Elements	Max midpoint error (°C)
10	1.25000000
20	0.31250000
40	0.07812500
80	0.01953125

The midpoint error is evaluated by interpolating the two FE nodal temperatures at each element midpoint and comparing that value with the exact quadratic solution. The error decreases by a factor of four when the element count doubles, consistent with O(h²) interpolation error. Constant properties, steady state, one-dimensional conduction and insulated lateral surfaces are assumed.

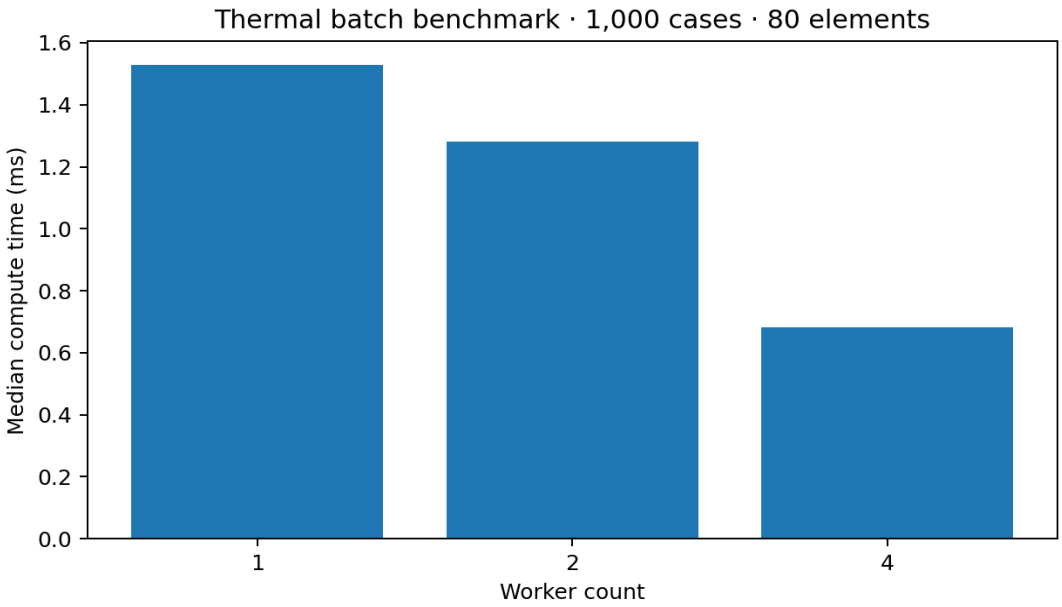


3. Geometry and data handling

The C++ geometry module reads ASCII STL vertex records, validates finite coordinates and reports triangle count and axis-aligned bounds. The CSV validator enforces the exact header node_id,x_m,y_m,z_m,temperature_C, rejects malformed/non-finite values and rejects empty datasets. The included STL is self-created sample geometry. The included nodal temperature CSV is synthetic and is not represented as an actual ANSYS export. Interactive 3D rendering and node inspection remain unimplemented.

4. Multicore execution

A fixed-seed generator creates independent cases with positive conductivity and heat generation. The serial path executes the batch in input order. Parallel paths use a bounded number of std::threads and an atomic work index; each case writes to its deterministic result slot. Five repetitions are measured and the median is reported. Speedup is T1/Tp and efficiency is speedup/p. The benchmark excludes compilation and file output; worker creation is included. Results are specific to the recorded Linux host.

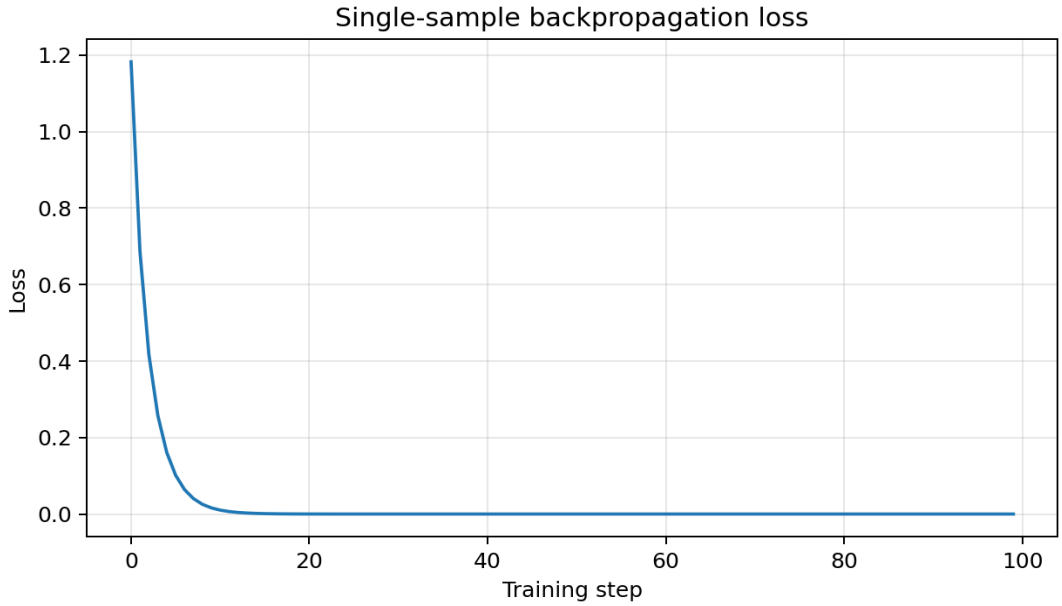


5. Embedded integration design

The embedded layer provides simulated device discovery and capability reporting for an STM32 Nucleo-class MCU, a Basys 3-class FPGA and a USB-JTAG adapter. It demonstrates connect/disconnect concepts, a telemetry message and a recoverable timeout. No physical device was connected. The interface table and block diagram in docs/embedded_interface.md describe illustrative 3.3 V assumptions; exact connector pinouts, voltage tolerance and power rules must be checked against the selected hardware revisions before any wiring. A JTAG adapter is a debug/programming transport, not a general-purpose compute board.

6. Backpropagation

For h=tanh(wx+b), y_hat=vh+c and J=0.5(y_hat–y)², the derivatives are dJ/dv=e·h, dJ/dc=e, dJ/dw=e·v·(1–h²)·x and dJ/db=e·v·(1–h²), where e=y_hat–y. The implementation computes all four gradients from the original parameter values before applying the simultaneous learning-rate-0.1 update. Central finite differences with epsilon 1e-6 agree with the analytic gradients within the required tolerance. See results/backprop_console.txt and the separate derivation appendix.



7. Verification and delivery status

The C++ project configures and builds with CMake on the Linux environment. CTest passes the integration test covering thermal boundary conditions and midpoint error, finite-difference gradients, all three mock devices, timeout behaviour, ASCII STL parsing, nodal CSV validation and serial/parallel agreement. Build and test evidence is in results/ and docs/requirement_matrix.md.

Known limitations: no integrated live GUI/backend transport, no interactive 3D viewer, no native Windows executable was built or clean-machine tested, no actual board was connected, and the handwritten appendix must be manually reproduced and scanned by the candidate. See docs/limitations.md. A Windows build script is supplied, but its output must be validated on Windows 11 x64 before submission.